

Credit Portfolio Risk Analysis

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Portfolios: Portfolio 1 and Portfolio 2

Course: Introduction to Credit Risk and Applications in Python

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1. Introduction

Credit portfolio risk quantifies potential losses arising from the simultaneous default of multiple obligors. Portfolio credit risk depends on both individual creditworthiness and the dependence structure among borrowers: when defaults are positively correlated, adverse macroeconomic conditions can trigger clustered losses far exceeding individual default probabilities in isolation (Vasicek, 1987). The Basel III Internal Ratings-Based (IRB) framework requires banks to estimate portfolio loss distributions and hold capital sufficient to absorb unexpected losses at high confidence levels (Basel Committee, 2006). This report analyzes the one-year credit risk of two simulated corporate portfolios, Portfolio 1 (28 Industry A obligors) and Portfolio 2 (24 Industry B obligors) implementing and comparing two models. The Gaussian One-Factor Model and the Beta-Bernoulli mixture model. Risk measures including Expected Loss (EL), Value at Risk (VaR 99%), and Expected Shortfall (ES 99%) are computed and interpreted.

2. Data

2.1 Dataset Description

Six datasets were used in the analysis, covering firm characteristics, portfolio exposures, market returns, and historical default information. The *entities_info.csv* file provides probability of default (PD), loss given default (LGD), industry, and region data for each firm, while the portfolio datasets contain exposure at default (EAD) information. Daily equity returns from *stock_returns.csv* and industry index returns from *industry_index_returns.csv* are used to estimate systematic sensitivity parameters (ρ). Historical default observations from *default_history_20y.csv* are used to compare analyst-estimated PDs with observed default experience.

2.2 Cleaning Procedures and Assumptions

Entity codes were standardized by removing whitespace and converting to uppercase to ensure consistent matching across all files. Industry labels with four inconsistent variants were mapped to canonical names. Date columns mixing ISO and European formats were parsed with mixed-format detection. Missing LGD (6 values) and PD (5 values) were imputed with sample medians of 0.471 and 2.67% respectively — chosen as robust estimates resistant to outliers, consistent with peer group credit quality and standard unsecured corporate recovery rates. Missing stock returns (2,242 rows) and rows with missing EAD (2 per portfolio) were dropped, as financial time-series cannot be reliably imputed and EAD is a required input for loss calculation. Missing default events (47 observations) were set to zero, under the assumption that an absent record indicates no observed default. Duplicates were removed keeping the first occurrence across all files.

After the cleaning process, the final modelling dataset contained the characteristics summarized in Table 1.

Table 1: Dataset Summary after Cleaning

Metric	Portfolio 1	Portfolio 2
Obligors	28	24

Metric	Portfolio 1	Portfolio 2
Total EAD (EUR M)	631.1	583.6
Mean PD (%)	2.034	3.311
PD range (%)	0.717 to 2.910	1.811 to 4.925
Mean LGD (%)	46.8	49.5
LGD range (%)	36.1 to 62.0	31.7 to 68.5
Total EL (EUR M)	5.7604	9.6328
Mean rho	0.441	0.540
rho range	0.410 to 0.495	0.480 to 0.579
Industry	Industry A	Industry B
Regions	Asia, Europe, N./S. America	Asia, Europe, N./S. America

3. Methodology

3.1 Literature Review

Vasicek (1987) derived the Gaussian One-Factor Model, providing a closed-form portfolio loss distribution under the assumption that firm asset returns follow a single-factor Gaussian model. The Basel Committee (2006) adopted this framework as the theoretical basis of the IRB approach, where banks estimate firm-level PD and LGD while a fixed asset correlation governs default dependence. Fitch Ratings (2008) examined the empirical validity of Basel II correlation assumptions by fitting Beta distributions to historical loss-rate data and inverting the IRB formula to obtain implied correlations, finding that regulatory values are conservative relative to historical experience. Merton (1974) establishes the theoretical foundation for using equity return correlations as proxies for asset correlations, since equity prices reflect firm asset value dynamics under the structural credit risk framework.

3.2 Gaussian One-Factor Model

The Gaussian One-Factor Model is a standard framework for portfolio credit risk measurement and forms the basis of the Basel Internal Ratings-Based (IRB) approach. The model captures default dependence through a common systematic factor that affects all obligors, while firm-specific shocks represent idiosyncratic risk.

$$X_i = \rho_i \cdot F + \sqrt{(1-\rho_i^2)} \cdot \varepsilon_i, F, \varepsilon_i \sim N(0,1)$$

where F is the common economic factor and ρ_i represents the systematic sensitivity of firm i . A default occurs when the latent variable falls below the threshold implied by the firm's probability of default (PD). Portfolio losses are calculated using simulated default outcomes together with exposure at default (EAD) and loss given default (LGD). The model was selected because it is widely used in practice and provides a transparent framework for measuring portfolio credit risk.

3.3 Beta-Bernoulli Model

The Beta-Bernoulli model treats the portfolio default probability as a random variable, allowing uncertainty in default rates to be incorporated directly into the loss simulation. Dependence between obligors arises through a shared portfolio-wide default probability rather than a common economic factor.

$$P \sim \text{Beta}(\alpha, \beta), Y_i | P \sim \text{Bernoulli}(P)$$

The parameters α and β are estimated using the portfolio mean PD and an assumed default correlation of 0.5% through the method of moments. The model was selected as an alternative dependence framework to compare with the Gaussian One-Factor Model and to evaluate the sensitivity of risk measures to different modelling assumptions.

3.4 Parameter Estimation

Probability of Default. PD is taken from `base_pd_hint` representing analyst-estimated one-year default probabilities. Validation against 20-year empirical default rates yields a Pearson correlation of 0.366 (Figure 2), confirming broad consistency with historical experience. The empirical PD is used for validation only, as many firms have zero historical defaults over 20 years, a known limitation of short records for low-default portfolios.

Systematic Sensitivity (ρ). The parameter ρ_i is estimated as the Pearson correlation between each firm's daily equity return and its industry index return, following Merton (1974). Estimates are clipped to $[0.1, 0.9]$ to ensure valid model inputs. The Basel Committee (2006) fixes ρ between 0.12 and 0.24 for regulatory IRB capital; our data-driven estimates are higher (mean 0.441 and 0.540 for Portfolios 1 and 2), consistent with Fitch Ratings (2008) who find that equity-return correlations tend to exceed asset correlations implied by historical loss data, reflecting leverage effects. **Simulation** uses 100,000 Monte Carlo scenarios (seed = 42) at 99% confidence, consistent with Basel III.

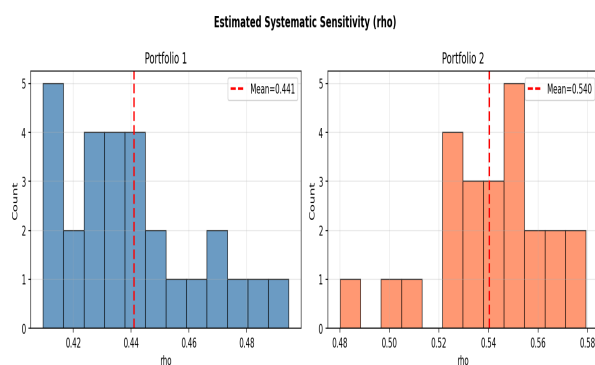


Figure 1: Distribution of estimated ρ by portfolio. Pearson correlation of firm equity return with industry index return.

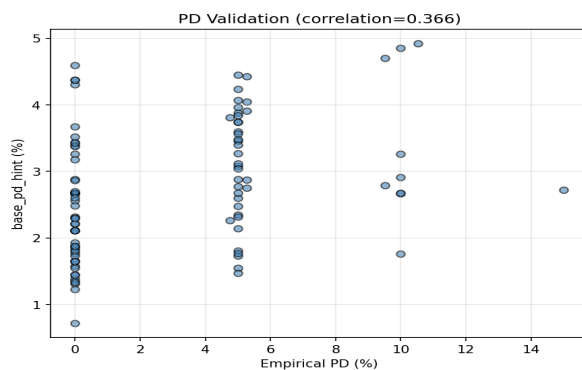


Figure 2: PD validation — `base_pd_hint` versus empirical PD from 20-year default history (correlation = 0.366).

4. Results and Discussion

4.1 Risk Measures

Table 2: Risk Measures by Portfolio and Model (100,000 simulations | 99% confidence | seed = 42)

Measure	P1 Gaussian	P1 Beta-Bernoulli	P2 Gaussian	P2 Beta-Bernoulli
EL	0.009128	0.009307	0.016506	0.015981
Std	0.017444	0.013956	0.030974	0.019966
VaR 99%	0.078888	0.056456	0.145372	0.080134
ES 99%	0.102181	0.067745	0.184874	0.093841

Gaussian ratios (P2 / P1): EL = 1.81 | VaR = 1.84 | ES = 1.81 Absolute EL: P1 = EUR 5.76M (0.913% of EAD) | P2 = EUR 9.63M (1.651% of EAD)

4.2 Portfolio Comparison

Portfolio 2 (Industry B) exhibits substantially higher credit risk across all measures and under both models. Under the Gaussian One-Factor Model, Portfolio 2 yields an Expected Loss of 1.65% of total EAD (EUR 9.63M) compared with 0.91% (EUR 5.76M) for Portfolio 1, a difference of 1.81 times. This is driven directly by the higher mean PD of Industry B firms (3.311% versus 2.034%) and their higher mean LGD (49.5% versus 46.8%), since the analytical formula $EL = \sum PD_i \cdot LGD_i \cdot w_i$ is a linear function of both parameters. The tail risk difference is more pronounced: VaR 99% for Portfolio 2 (14.54% of EAD) exceeds Portfolio 1 (7.89%) by 1.84 times, and ES 99% (18.49% versus 10.22%) by 1.81 times. The gap between EL and VaR of approximately 8 to 9 times reflects the right-skewed nature of credit loss distributions: most scenarios produce near-zero losses while rare adverse conditions generate disproportionately large losses. Portfolio 2 firms exhibit higher systematic sensitivity (mean $\rho = 0.540$ versus 0.441), which strengthens the link to the common economic factor and widens the tail without affecting EL, since the EL formula is independent of ρ (Vasicek, 1987). ES to VaR ratios of 1.30 and 1.27 for Portfolios 1 and 2 respectively confirm heavy-tailed distributions in both cases.

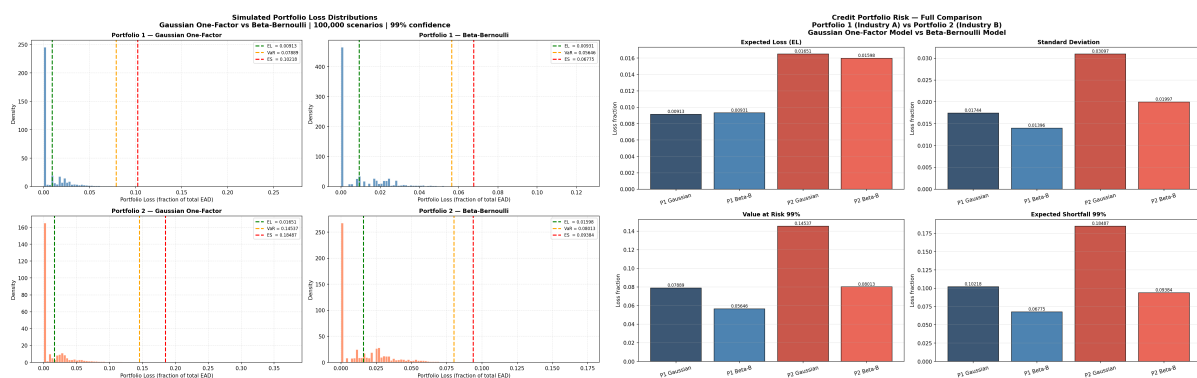


Figure 3: Simulated loss distributions — Gaussian (top) vs Beta-Bernoulli (bottom). Green = EL | Orange = VaR 99% | Red = ES 99%.

Figure 4: Risk measure comparison across both portfolios and both models.

4.3 Model Comparison

Both models yield closely aligned EL estimates for each portfolio — Portfolio 1: 0.9128% (Gaussian) versus 0.9307% (Beta-Bernoulli); Portfolio 2: 1.6506% versus 1.5981% confirming that EL is invariant to the dependence structure and depends solely on PD, LGD, and EAD. The models diverge substantially in tail risk: Gaussian VaR 99%

exceeds Beta-Bernoulli VaR 99% by 40% for Portfolio 1 and 82% for Portfolio 2, with ES differences of 51% and 97% respectively. The Gaussian model generates higher tail risk because the estimated rho values (mean 0.441 and 0.540) translate into strong systematic dependence through the common factor, while the Beta-Bernoulli model parameterizes dependence through a default correlation of only 0.5%. This divergence illustrates model risk in credit portfolio measurement: the choice of dependence structure can alter capital requirement estimates by a factor approaching two. Consistent with Fitch Ratings (2008), our equity-return based correlations exceed values implied by historical loss data, reflecting leverage effects and the conservative calibration of the Basel regulatory framework.

5. Conclusion

This report quantified the one-year credit risk of two simulated corporate portfolios using the Gaussian One-Factor Model and the Beta-Bernoulli model applied to systematically cleaned and merged datasets. Three principal findings emerge. First, Portfolio 2 (Industry B) exhibits materially higher risk than Portfolio 1 (Industry A) across all measures, EL 1.81 times higher, VaR 99% 1.84 times higher, and ES 99% 1.81 times higher driven by elevated PD, LGD, and systematic sensitivity parameters. Second, both models produce consistent EL estimates but diverge by 40 to 97% in tail risk measures, illustrating that capital requirements are sensitive to the choice of dependence model. Third, ES to VaR ratios of 1.27 to 1.30 confirm heavy-tailed loss distributions, underscoring ES as the preferred regulatory risk measure over VaR. Limitations include the short rho estimation window of approximately two years, which may understate stress-period correlations (Fitch Ratings, 2008), and reliance on analyst PD estimates that may not fully reflect current market conditions. Future work could incorporate stress-tested correlation scenarios and time-varying PD models calibrated to credit spreads.

6. References

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